



TEST & MEASUREMENT

modern EW systems with better resolution and greater frequency agility.

A&D industry is focused on delivering mission-critical systems on time with the latest technologies incorporated. This means that the relevant players must come up with innovative products in reduced time, apart from producing reliable, fail-safe products. This contradiction of providing more functionality and reliability in a limited time frame is a big challenge for this industry.

T&M equipment are expected to have low latency and tight control loops. Defence T&M equipment should be rugged and meet MIL standards. Instruments should be portable and battery-operated, so that these can be taken into field if required. Computation techniques such as filtering, fast fourier transforms (FFTs), digital down-conversion and channelisation are implemented in the hardware to reduce the amount of data processed by the CPU. T&M equipment for defence should be usable in highly humid conditions, extreme temperatures, be resistant to shock vibrations and other special conditions.

Instrumentation targeted at these industry sectors needs to be highly accurate and deliver consistent results. Test products that are optimised for such applications also have to be flexible in many cases. Mohit Chauhan, pre-sales manager at MAKSAT Technologies, says that, at his company they utilise RF spectrum analysers for most testing requirements, such as immersion tests, local area network (LAN) tests, link performance tests and shock tests, among others.

Modular instruments enable to prototype electronics system by connecting input/output with software. For example, advanced radar systems can be prototyped by RF signal analysers and digitisers. User-programmable FPGA connectivity allows quicker advanced algorithm deployment in prototypes and faster proof-of-concept validation. With software-designed instrumentation, data can be preprocessed in the hardware, reducing CPU load significantly.

Spectrum analysers, vector network analysers, signal generators, transponder test sets, radio altimeters, logic analysers, oscilloscopes and power meters are some of the most demanding test instruments for D&A industry. All Rohde & Schwarz signal and spectrum analysers, from basic handheld models to benchtop instruments, support a wide range of frequencies from 10Hz to 85GHz. Keysight has Advanced X series, which supports frequency range from 2Hz to 110GHz.

Shriram Shinde, RF application engineer at CoreL Technologies,

Higher bandwidth devices.

There is a shift towards RADAR and other EW technologies that have more signal-processing capabilities (such as cognitive RADARs). These technologies require T&M equipment capable of processing large RF bandwidths. There is an expected demand for such products, which can provide high bandwidth and inline signal processing for high bandwidth data.

Large-scale hardware in the loop (HIL). This area has not been very popular in the past but, with more electronics and embedded

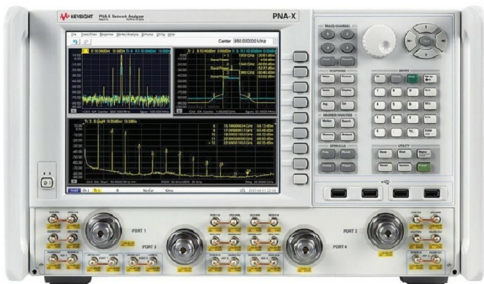


Fig. 2: Keysight Technologies integrates spectrum analyser capability into its PNA vector network analysers (Credit: www.rfglobalnet.com)

says, T&M equipment play a major role in ensuring reliability of RF communication in defence. High accuracy is the most important feature of T&M equipment for defence applications. There are various T&Ms performed under test and evaluation processes. Spectrum analysers are the most used T&M equipment for testing frequency domain characteristics.

One of the most commonly faced challenge with T&M equipment is their power-handling capabilities; these do not provide the facility for high power measurements. Power network analysers are being developed to overcome this challenge.

Future opportunities

Kulshrestha highlights two opportunities in T&M equipment:

software coming onboard A&D products, HIL is certainly picking up pace. One key aspect to watch out for is that, these HIL systems have different needs than traditional automotive HILs. These systems will need to be more reliable and highly scalable, apart from being able to incorporate tight synchronisation on a very large scale of sensors and actuators.

Calibration is an important aspect for T&M equipment. Inaccuracy in T&M equipment leads to inaccurate results and complications. Because of this, military industry is also focussing on this aspect of testing. ISO/IEC 17025:2005 certification on all testing equipment has become an absolute must and is considered as a standard for D&A industry as well. **EFY**